

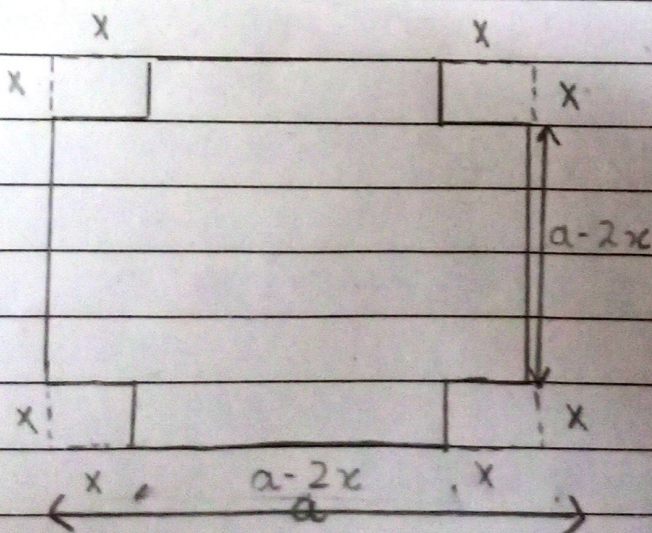
Date: _____

COMPLEX ENGINEERING PROBLEM

STATEMENT

A square sheet of cardboard with each side a centimeters is to be used to make an open-top box by cutting a small square of cardboard from each of the four corners and bending up the sides. What is the side length of the small squares if the box is to have as large a volume as possible?

SOLUTION



Let, the side length of the small square be x cm. The side length of the open box is $(a-2x)$ cm and the height is x cm. Here, a is constant and x is the variable.

The volume V cm³ of the box is given by

$$V(x) = x(a-2x)^2 = 4x^3 - 4ax^2 + a^2x$$

Maximizing the Volume

Differentiate the volume function

$$\frac{dv}{dx} = 12x^2 - 8ax + a^2$$

Setting the Derivative to zero And Solve for

$$\frac{dv}{dx} = a^2 - 8ax + 12x^2$$

$$a^2 - 8ax + 12x^2 = 0$$

$$x = \frac{-(-8a) \pm \sqrt{(-8a)^2 - 4(12)a^2}}{2(12)}$$

$$x = \frac{8a \pm \sqrt{64a^2 - 48a^2}}{24}$$

$$x = \frac{8a \pm \sqrt{16a^2}}{24}$$

$$x = \frac{8a + 4a}{24}$$

$$x = \frac{12a}{24}$$

$$x = \frac{4a}{24}$$

$$x = \frac{a}{2}$$

$$x = \frac{a}{6}$$

determining Which Value Of x Maximizes

we will use Second Derivative test to figure out

$$\frac{V}{x^2} = 24x - 8a$$

$$x = a/6$$

$$\frac{V}{x^2} = 24\left(\frac{a}{6}\right) - 8a$$

$$\frac{V}{x^2} = 4a - 8a$$

$$\frac{dV}{dx^2} = -4a$$

Since, $-4a < 0$, which means that

$x = a/6$ is maximum point. Therefore, the length $x = a/6$ will give maximum volume.

At $x = a/2$

If $x = a/2$, the length of each side of the box (after cutting out the corners) becomes;

$$a - 2x \Rightarrow a - 2\left(\frac{a}{2}\right) \Rightarrow a - a \Rightarrow 0$$

This means the box has no base area because the length of each side is zero which makes volume zero.

Thus, when $x = a/2$, there is no actual box.

Maximum Volume Calculation

Now, we know the max. volume occurs at $x = a/6$; substitute into volume formula

$$V(x) = x(a - 2x)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(a - 2 \times \frac{a}{6}\right)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(\frac{2a}{3}\right)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(\frac{4a^2}{9}\right)$$

$$V\left(\frac{a}{6}\right) = \frac{2a^3}{27}$$

So, the max. volume is $V\left(\frac{a}{6}\right) = \frac{2a^3}{27}$



CONCLUSION

The maximum volume occurs when $x = \frac{a}{6}$ and the maximum volume is $\frac{2a^3}{27}$ so the easiest way to solve this problem is to cut the small squares of size $x = \frac{a}{6}$ from each corner. This will give you an open-top box with maximum volume possible.

REAL WORLD USE

Consider you are a packaging company and you need to create a box using the least amount of material while still maximizing its capacity. This could be useful in minimizing waste, lowering cost and maximizing storage space.

FIELD OF THE PROBLEM

The problem uses differentiation and maximization methods from Calculus and applies them to real world fields such as manufacturing and engineering design. This shows how Calculus plays a key role in solving practical problems.